Decentralized Identifier System for Software Copyright Transfer and License Management

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***Abstract*—This paper presents a distributed identifier (DID) system that offers transparent and traceable outsourcing support platform for software development. The proposed system is designed to ensure the privacy of contracting parties during the copyright transfer and license contract process, thereby creating a fair and transparent outsourcing market. By leveraging a DID system, our approach ensures that no single entity controls the information, which increases the overall security and reliability of the system. Also, it reduces corporate losses due to non-compliance with copyright and license contracts.**

***Index Terms*—DID, Copyright, License, Verifiable Credentials**

I. Introduction

With software development becoming more intricate, many organizations are turning to outsourcing as a means to reduce expenses and access specialized features [1]. Nevertheless, outsourcing software development comes with significant challenges such as maintaining high software quality and avoiding conflicts arising from copyright and licensing contracts. As transactions to manage licenses and transfer software copyright are expected to occur in the metaverse, similar to the physical world, there is a need for transparency and traceability to create a secure and open outsourcing market in the metaverse environment.

To achieve this goal, this paper suggests a decentralized identifier (DID) system that ensures privacy during the process of software copyright transfer and license management. Specifically, our approach uses Restructured Verifiable Credentials (VC) and Software Bill of Materials (SBOM) to establish a secure and transparent outsourcing market.

This paper is as follows: Section II highlights the potential benefits of the Restructured VC and SBOM for software development industry. Section III describes our proposed DID system. Section IV summarizes our findings and concludes the paper.

II. Related Work

A. Restructured Verifiable Credentials

The use of DID to confirm personal details of individuals can potentially jeopardize the confidentiality of their identity-related information, thus putting the data subjects at risk of exposure. To minimize this risk, a Zero-Knowledge Proof (ZKP) is used to ensure anonymity and pseudonymization. But, there is still a chance of identifying the data subject. To address this challenge, a new concept was introduced called Restructured VC. This approach involves generating a ZKP_{VC} through a ZKP with the certifier and a VC_R generated using the verifier's public DID identifier [2]. This ensures that there is no correlation between the identity information and the attestant, making it possible to share external data without the attestant's permission.

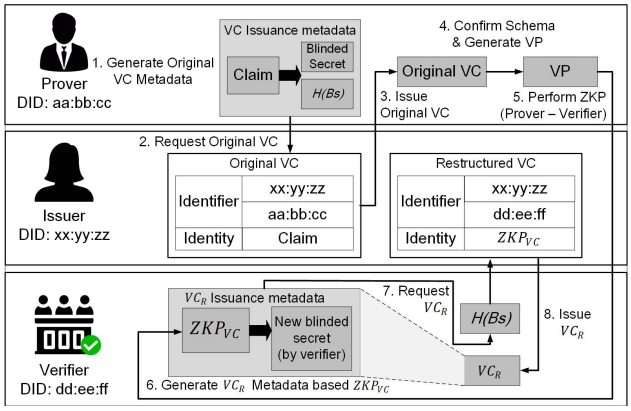


Fig. 1. Generation Process of the Restructured VC

Fig. 1. illustrates the process of generating the Restructured VC. The verification process begins with the Prover generating the Original VC Metadata. Next, the Prover requests the Original VC from the Issuer and generates a VP from the Original VC using the Confirm Schema. The Prover then performs the ZKP and delivers it to the Verifier. The Verifier obtains new metadata from the ZKP value generated from the VC. Finally, when the Verifier requests a restored VC from the Issuer, the Issuer issues the VC_R. This step-by-step process ensures that the VC is authentic and has not been tampered with, providing a secure and reliable verification mechanism for all parties involved.

B. Software Bill of Materials

The SBOM is a structured descriptive list that outlines the details and dependencies of the components that make

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up a software system. The Software Package Data Exchange (SPDX), a format widely adopted SBOM, aims to specify software components and license information, making it easier to manage open-source licenses and utilize the SBOM. Creating SPDX SBOM can be automated with automatic generation tools. By automatically generating SBOM, developers can effortlessly handle license information and determine the constituent elements of the software, while also enhancing the transparency and security of the supply chain.

III. Decentralized Identifier System for Software Copyright Transfer and License Management

Fig. 2. shows the architecture of our proposed DID system for software copyright transfer and license management. By leveraging advanced technologies such as Restructured VC and ZKP, our system ensures that no single entity controls the information and that the identity attributes of data subjects are protected. This provides a more secure and reliable platform for managing software licenses, ensuring compliance with industry regulations and minimizing corporate losses due to non-compliance. By adopting our proposed decentralized identifier system, organizations can improve the quality of their outsourced software development and reduce corporate losses due to non-compliance with copyright and license contracts. Through the proposed system, the Copyright Commission can deliver the VC containing the SBOM recording copyright and license information of software. Table I shows the metadata expected to be used for the VC in our proposed system.

First, the VC is issued in the proposed system using VC metadata. Contracting parties generate a DID value with Restructured VC through the ZKP to minimize the exposure

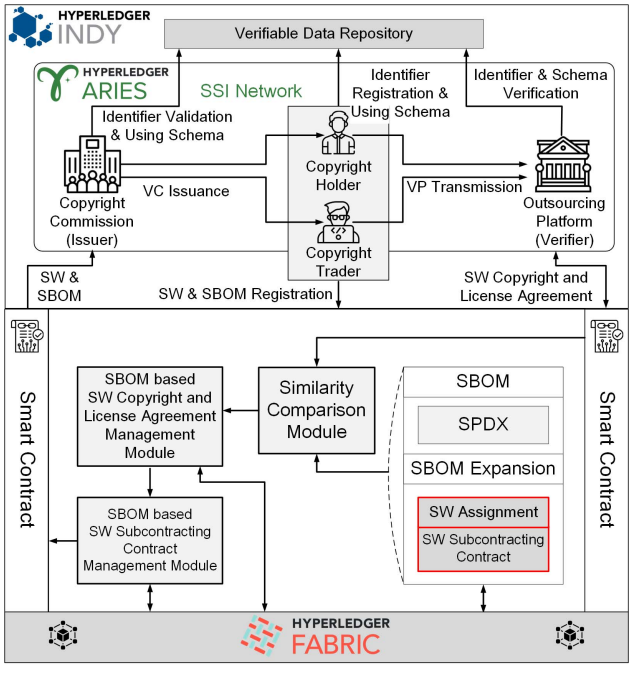


Fig. 2. Decentralized Identifier System for Software Copyright Transfer and License Management

TABLE I
VC Metadata

Category	Metadata
Software Basic Information	- SW name
	- SW creation date
	- SW announcement data
Personal Information	- Name (Corporate name)
	- Resident registration number (Corporate registration number)
	- Business registration number
	- Address
	- Email
	- Homepage
SBOM	- SPDX
	- SPDX Expansion

of their identity attribute information. Identity authentication is performed through the Aries Self-Sovereign Identity (SSI) Network, with the Schema and Identifier values of the certificate registered in the Verifiable Data Repository to manage DID information. Once identity verification is completed and the contract is executed, software development can proceed, and an SBOM is generated based on the terms of the contract. Smart contracts are used to verify the results of the software development process, and the Similarity Comparison Module is used to compare the SBOM and software generated by the system with previously registered SBOM data. Once the system passes through the Similarity Comparison Module, the SBOM-based SW Copyright and License Agreement Management Module is triggered, and the test results are stored on the blockchain. Finally, the SBOM-based SW Subcontracting Contract Management Module is responsible for updating the copyright and license information of the software according to the terms of the contract and transmitting the newly updated information to the Copyright Commission.

IV. Conclusion

The proposed system in this paper utilizes DID technology to guarantee privacy during software copyright transfer and license contracts. This system offers a pathway to a decentralized system-based outsourcing support platform that ensures transparency and traceability, ultimately reducing the possibility of corporate losses during software development outsourcing in both the physical world and the metaverse.

Acknowledgment

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